RCD-IIUM: A Comprehensive Malaysian Road Crack Dataset for Infrastructure Analysis

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***Abstract*— In rapidly urbanizing regions, maintaining road infrastructure integrity is a critical challenge due to increasing vehicular stress and environmental factors. This study introduces the Road Crack Dataset-International Islamic University Malaysia (RCD-IIUM), designed to enhance road pavement infrastructure management in Malaysia. Employing advanced data collection technologies, including high-resolution digital imaging, the dataset captures detailed anomalies in road surfaces, laying the groundwork for robust infrastructure analysis. The utility and efficacy of the RCD-IIUM dataset were evaluated through the deployment of three deep learning models: Customized YOLOv7, YOLOv8X-SEG, and an Advanced Hybrid Deep Learning Model. These models were tested for their ability to detect and classify road cracks using metrics such as precision, recall, F1-score, and overall accuracy. Results indicated that the YOLOv8X-SEG model outperformed others, demonstrating higher accuracy of 90% and F1-score of 95%. The Customized YOLOv7 model achieved a precision of 93%, recall of 91.58%, and overall accuracy of 88%. The Advanced Hybrid Deep Learning Model achieved a precision of 88%, recall of 89%, F1-score of 88.5%, and overall accuracy of 85%, further validating the robustness of the dataset. The dataset not only bolsters road pavement maintenance strategies but also supports data-driven decision-making for urban planning and policymaking.**

Keywords*— Deep Learning, Road Infrastructure, Pavement Management System, RCD-IIUM, Pavement Maintenance.

I. INTRODUCTION

The resilience of road infrastructure is paramount to Malaysia's economic vitality and societal well-being, facilitating the seamless flow of goods, services, and people [1]. However, road networks in Malaysia frequently suffer from degradation due to environmental stresses, intense vehicular load, and often insufficient preemptive maintenance actions. Among various forms of road pavement distress, cracks are particularly concerning as they are early indicators of potential structural failures that could lead to increased operational costs and pose safety risks [2], [3].

Effective management and maintenance of road pavement infrastructure necessitates high-quality, comprehensive data that accurately reflects current conditions and can forecast future deterioration [4]. In Malaysia, there is a conspicuous absence of a centralized, systematic road crack dataset that encompasses detailed, actionable information suitable for extensive analytical applications. This deficiency severely limits the precision and effectiveness of predictive maintenance strategies and the optimization of repair operations [5], [6].

To bridge this gap, this paper introduces the Road Crack Dataset-International Islamic University Malaysia (RCD-IIUM). This dataset is specifically designed to provide a repository of road surface anomalies from the regions of Selangor and Kuala Lumpur, with the methodology and framework developed being scalable to a national level. The data collection process employed in the RCD-IIUM involved high-resolution digital imaging. This approach not only ensures the capture of detailed, georeferenced imagery of pavement cracks but also facilitates the layering of additional data streams, enhancing the dataset's multidimensionality and utility.

A significant feature of the RCD-IIUM dataset is its compatibility with advanced Artificial Intelligence (AI) and Machine Learning (ML) methodologies. These technologies are at the forefront of transforming static data into dynamic models that predict road degradation, optimize maintenance schedules, and streamline resource allocation. By applying algorithms capable of analyzing complex patterns and predicting future outcomes, the dataset can significantly improve the responsiveness and efficiency of road pavement maintenance programs. For instance, convolutional neural networks (CNNs) and recurrent neural networks (RNNs) can be employed to perform image classification and time-series forecasting, respectively, to identify crack patterns and predict their progression over time [7], [8], [9].

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The effectiveness of a machine learning model is significantly influenced by the quality and suitability of the data used in its development. Therefore, choosing an appropriate dataset is crucial. Numerous pavement crack datasets have been utilized by researchers to develop models for road crack detection and classification. TABLE I provides an overview of the datasets included in this framework.

TABLE I. LIST OF DATASETS FOR PAVEMENT CRACK ANALYSIS

Dataset	Image Sample Count	Image Resolution	Image Format	Data Enhancement
Concrete Crack Images for Classification [10]	40000	227 x 227 pixels	Colour	Not Applicable
RDD2020: An Image Dataset for Smartphone-based Road Damage Detection and Classification [11]	26336	600 x 600 pixels	Colour	Not Applicable
Asphalt Crack Dataset [12]	400	448 x 448 pixels	Colour	Not Applicable
EdmCrack600: a Pixel-level annotated dataset for crack identification [13]	600	256 x 256 pixels	Colour	Not Applicable
Crack Detection: Image Classification	15168	Not Available	Colour	Not Applicable
Cracks-and-Potholes-in-Road-Images-Dataset [14]	2235	2560 x 1440 pixels	Colour	Not Applicable
CFD [15]	118	320 x 480 pixels	Colour	Not Applicable
AigleRN database [16]	38	Not Available	Greyscale	Not Applicable
CRACK500	500	2000 x 1500 pixels	Colour	Applicable
German Asphalt Pavement Distress (GAPs) Dataset [17]	1969	1920 x 1080 pixels	Greyscale	Not Applicable
Cracktree200 [18]	206	800 x 600 pixels	Colour	Not Applicable

The objectives of this study are multifaceted: to delineate the comprehensive data collection and validation methods employed in the creation of the RCD-IIUM dataset, to explore its application in developing robust AI and ML models for infrastructure management, and to illustrate its potential impact on national road maintenance strategies. By leveraging detailed empirical data and cutting-edge analytical tools, this research aims to establish a new standard for infrastructure management, promoting a shift from reactive to proactive maintenance practices across Malaysia's road networks. This will not only enhance road safety but also extend the lifespan of road assets, ensuring sustainable infrastructure development aligned with Malaysia's growth ambitions.

This paper explores the development, utilization, and potential enhancements of the Road Crack Dataset from International Islamic University Malaysia (RCD-IIUM). It begins with a detailed Methodology section that outlines the data acquisition techniques such as high-resolution digital imaging and geospatial mapping used to compile the dataset. The Dataset Description section offers an overview of the dataset's characteristics, including types, format, and coverage. Analysis and Findings assess the dataset's efficacy through three advanced deep learning models—Customized YOLOv7, YOLOv8X-SEG, and an Advanced Hybrid Deep Learning Model—highlighting their performance on key metrics like precision, recall, and accuracy. Accessibility of the Dataset addresses the dataset's availability and the required citation for its use, aiming to foster collaborative research and practical applications in road infrastructure management. The Conclusion and Future Scope section consolidates the study's contributions and outlines future research directions, focusing on enhancing model accuracy, expanding dataset coverage, and integrating models into real-time monitoring systems to extend the impact of this research on a broader scale.

II. METHODOLOGY

A. Acquisition and Setup

To achieve a robust dataset that underpins the efficacy of deep learning models, it is crucial to collect data that accurately represents real-world conditions [19], [20]. For this purpose, the research employed a GoPro Hero 8 camera mounted on an inspection vehicle namely Perodua Viva specifically customized for this project. This setup aimed to emulate the typical viewing angle and height experienced by a vehicle's onboard camera, ensuring that the dataset reflects real driving scenarios as closely as possible. The methodological diagram is shown in Fig. 1.

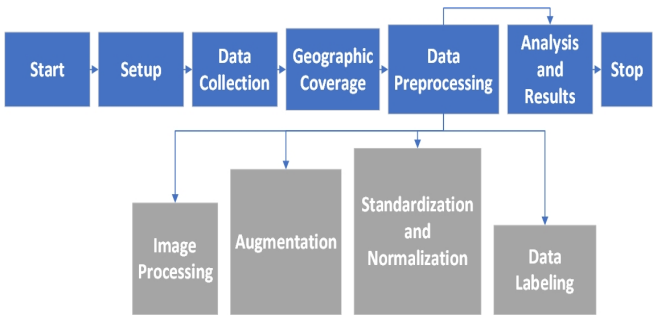


Fig. 1. Methodological Diagram

The camera was configured to cover a road width of 3.1 meters, aligning with the average width of a single traffic lane in Malaysia. This specific setup allows the collection of consistent and applicable data across different locations, ensuring that the dataset is not only comprehensive but also standardized across varying road conditions. The precise camera specifications, including the frame rate, resolution, and lens type, are crucial for capturing high-quality images necessary for detailed analysis and are summarized in TABLE II. The outcomes of various calibration setups, designed to optimize the data accuracy, are detailed in TABLE III.

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TABLE II. SPECIFICATIONS OF THE CAMERA CONFIGURATION

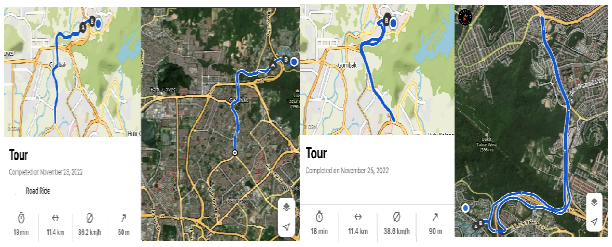
Attribute	Details
Camera Model	GoPro Hero 8
Mounting Hardware	GoPro Rod Mount
Operating Mode	Custom Video Setting
Image Resolution	1080p (1920×1080)
Frame Rate Options	24 fps, 60 fps
Lens Type	Linear Field of View
Video Bit Rate	Standard Quality (45 Mbps)
Minimum ISO Setting	100

TABLE III. OUTCOMES OF CAMERA CALIBRATION

Calibration Setup	Road Coverage Width	Camera Orientation to Ground	Proximity to Marked Road Segment	Installation Height
Setup 1	3.1 meters	90° ± 35°	1.1 meters	1.30 meters
Setup 2	3.1 meters	90°	Directly above midpoint	1.6 meters

B. Geographic Coverage

The geographic scope of data collection includes road networks around Selangor and Kuala Lumpur. These areas were chosen due to their significant traffic volumes and diverse road conditions, which provide a broad spectrum of data on road wear and tear. Detailed GPS mapping of the data collection routes was incorporated into the study to ensure clarity and reproducibility of the research. Fig. 2(a) and Fig. 2(b) showcase these maps, offering future researchers the context needed to understand and potentially replicate the study in other regions or compare different datasets.



(a) (b)

Fig. 2. (a) GPS road Data from Selangor, (b) GPS road Data from Kuala Lumpur

C. Data Preprocessing

1) Image Processing

Upon acquiring the video footage, the data were segmented into frames, transforming the continuous visual stream into manageable, static images for analysis as shown in Fig. 3. This segmentation is critical as it converts dynamic real-time data into a format that can be effectively used in training machine learning models. Each frame was resized to a uniform resolution of 640x640 pixels, ensuring consistency across all data inputs, which is essential for maintaining computational efficiency and algorithm accuracy.

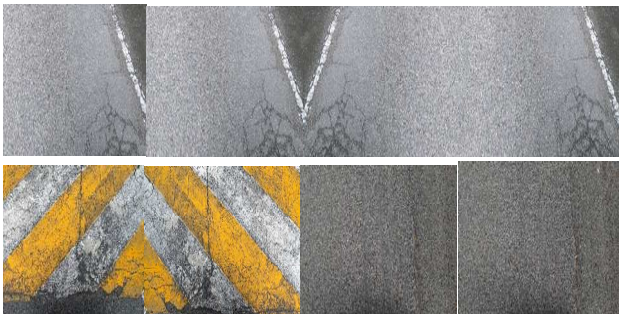


Fig. 3. Sample Images

2) Augmentation Techniques

To enhance the robustness of the dataset, various image augmentation techniques were applied. Horizontal and vertical flips were introduced with a predetermined probability, effectively doubling the dataset's size and providing varied perspectives of the same road conditions. Additionally, adjustments to image brightness and exposure were made within a range of +20% to -20% along with blurring within range of 2px to 4.5px to simulate different times of day and weather conditions, thus preparing the model to operate under diverse environmental settings. These preprocessing steps are designed to make the dataset as representative and comprehensive as possible, thereby improving the model's ability to generalize to new, unseen data.

3) Data Standardization and Normalization

In preparing the dataset for use in our models, we implemented data standardization and normalization, focusing on enhancing the dataset's applicability to real-world scenarios. This approach included maintaining a balance between processed and noised image data. While standardization aligns all images to a consistent format and scale—such as resizing all frames to the same pixel dimensions—normalization adjusts image data values to a specific range, ensuring uniform input to deep learning algorithms.

Crucially, to mimic real-world imperfections and variances, we intentionally included images with various noise levels alongside the processed data. This strategy involves incorporating image data that has not been corrected for brightness, contrast, or noise, thereby preserving the natural variability encountered in real road environments. This mixture of pristine and noisy data is vital for training robust models capable of performing accurately under diverse and unpredictable real-world conditions.

By combining processed images with those retaining certain 'real-world' imperfections, we ensure that our dataset better prepares the models for the complexities of actual road conditions. This preparation is crucial for developing AI and ML systems that can reliably interpret and react to the wide range of visual information they might encounter in practical applications, from clear, sunny days to overcast, poorly lit conditions.

D. Data Labeling and Utilization

1) Labeling Strategy

Accurate data labeling is fundamental in supervised learning scenarios. Using the Roboflow data annotation tool [21], each image frame was labeled to indicate the presence

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and type of road cracks: alligator, longitudinal, and transverse as shown in Fig. 4. This detailed categorization allows the model to not only detect the presence of cracks but also classify them, which is crucial for assessing the severity and required interventions.

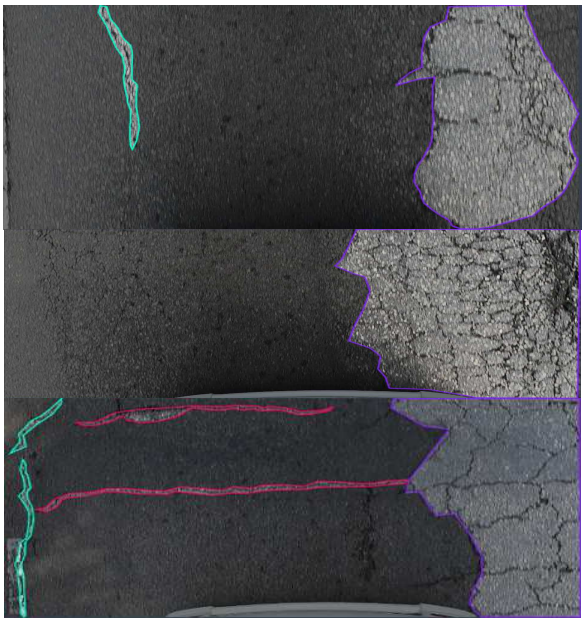


Fig. 4. Examples of annotated image frames: Purple showing Alligator crack, Pink showing Transverse crack and Blue showing Longitudinal crack.

2) Data Split

The dataset was divided into training, validation, and testing segments, as shown in TABLE IV. This separation is crucial for training the models comprehensively and assessing their performance under controlled conditions. The data split was approximately 85% for training, 10% for validation, and 5% for testing. This distribution ensures that the model can learn from a varied set of examples and is tested against unbiased data, providing a reliable measure of its predictive power and practical utility.

TABLE IV. SUMMARY OF DATA DISTRIBUTION FOR TRAINING, VALIDATION, AND TESTING OF ROAD CRACK CLASSES

Class	Label	Training Images	Validation Images	Testing Images	Total
Alligator Cracks	crack-alligator	760	80	40	880
Longitudinal Cracks	crack-long	760	80	40	880
Transverse Cracks	crack-trans	380	47	23	450
Total		**1900**	**207**	**103**	**2210**

Through this comprehensive methodology, the RCD-IIUM dataset provides a solid foundation for applying advanced AI and ML techniques to road maintenance and infrastructure management, potentially transforming how such tasks are approached and executed across Malaysia.

III. Dataset Description

A. Characteristics of the RCD-IIUM Dataset

The RCD-IIUM dataset has been designed to offer high-resolution, comprehensive data on road pavement surface conditions, specifically focusing on road cracks. This dataset is characterized by several key attributes:

- *Data Types and Format:* The dataset primarily consists of high-resolution video footage and derived still images captured using a GoPro Hero 8 camera. These images are stored in a standard digital format, ensuring they are readily accessible and usable for analysis.
- *Diverse Data:* The data collection for the RCD-IIUM dataset was conducted across extensive road networks in Selangor and Kuala Lumpur. The focus was on capturing comprehensive road surface images, each spanning a full width of 3.1 meters—typical of a standard road lane. To ensure the accuracy and consistency of these images, the camera was meticulously set at a height of 1.6 meters, positioned parallel to the road surface.
- *Variables Recorded:* The primary variables include crack type (alligator, longitudinal, transverse), location, and environmental conditions (lighting and weather conditions at the time of capture).

B. Visual Examples and Sample Data Entries

To ensure clarity and utility of the data, visual documentation of the setup was recorded. Fig. 5 presents the data collection setup, demonstrating the practical application of the configured camera in real-world conditions. This visual aid helps contextualize the technical descriptions and provides a clear depiction of the experimental setup used during the data acquisition phase.



Fig. 5. Data Collection Setup

Additionally, sample data entries include annotated images showcasing various types of road cracks, annotated with precise labels to aid in the identification and classification processes during machine learning model training.

IV. Analysis And Findings

A. Overview

The experiments were conducted on an HP Pavilion 15-bc408tx system equipped with an Intel Core i7-8750H processor, 8 GB of DDR4 memory, and a 1 TB hard disk drive. The system also utilized an NVIDIA GeForce GTX

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1050 with 4GB VRAM for onboard graphics and an NVIDIA GeForce RTX 4080 for remote graphics processing. The development platforms used in this study were Anaconda and Google Colab. The machine learning frameworks employed included PyTorch for the YOLO models (Customized YOLOv7 and YOLOv8X-SEG) and TensorFlow for the Advanced Hybrid Deep Learning Model.

B. Model Descriptions and Comparative Performance

1) Model Overview

Three customized models were employed to evaluate the dataset:

- *Customized YOLOv7:* Optimized for rapid detection and broad classification of road cracks.
- *Customized YOLOv8X-SEG:* Focused on instance segmentation, providing precise localization and detailed mapping of cracks.
- *Advanced Hybrid Deep Learning Model:* Integrates deep learning with traditional computational methods to improve detection accuracy across diverse environmental conditions.

2) Evaluation Metrics

Each model was tested using standard performance metrics including precision, recall, F1-score, and overall accuracy. These metrics are essential for understanding each model's ability to correctly identify and classify various crack types within the dataset as shown in TABLE V.

TABLE V. COMPARATIVE PERFORMANCE METRICS

Model	Precision	Recall	F1-Score	Overall Accuracy
Customized YOLOv7	93%	91.58%	92%	88%
Customized YOLOv8X-SEG	90%	89%	95%	90%
Advanced Hybrid Deep Learning Model	88%	89%	88.5%	90%

3) Visualization

The effectiveness of each model is illustrated through visual outputs in Fig. 6, Fig. 7 and Fig. 8, demonstrating the models' capabilities in distinguishing between true and predicted classifications of crack types.

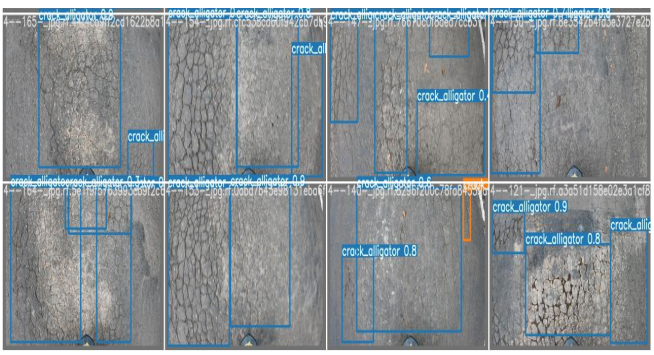


Fig. 6. Customized YOLOv7 Model Predictions

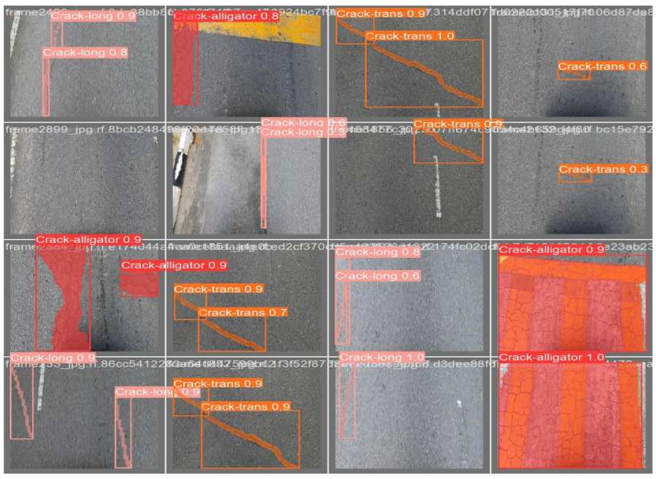


Fig. 7. Customized *YOLOv8X-SEG* Model Predictions

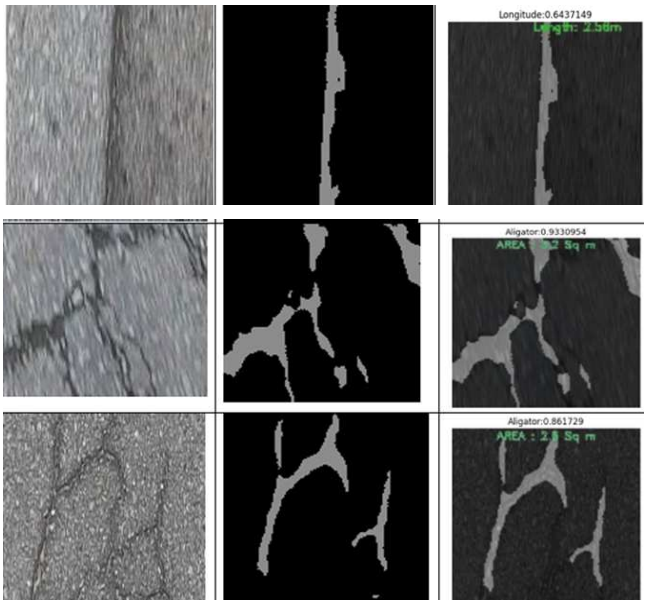


Fig. 8. Advanced Hybrid Deep Learning Model Predictions

4) Practical Implications of the Findings

The application of these models using the RCD-IIUM dataset highlights several practical implications for infrastructure management and maintenance:

- *Enhanced Detection Capabilities:* The high accuracy and precision demonstrated by the models are indicative of their potential to significantly improve crack detection, essential for timely interventions.
- *Optimized Resource Allocation:* With precise classification capabilities, maintenance efforts can be more effectively directed to areas with the most severe damage, optimizing both costs and labor allocation.
- *Data-Driven Maintenance Decisions*: The depth and accuracy of data provided by these models can substantially enhance decision-making processes, enabling more strategic planning in road maintenance schedules and infrastructure investments.

5) Benchmarking Comparison

To effectively gauge the performance of our models against industry standards, we performed a benchmarking

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comparison presented in TABLE VI. This analysis compares our models—Customized YOLOv7, YOLOv8X-SEG, and the Advanced Hybrid Deep Learning Model—with notable benchmarks from the literature. This comparative study helps in highlighting the effectiveness of our models in the realm of road crack detection technology.

TABLE VI. BENCHMARKING

Model	Recall	Precision	F1-Score	Overall Accuracy	Notes
Customized YOLOv7	91.58%	93%	92%	88%	This study
Customized YOLOv8X-SEG	89%	90%	95%	90%	This study
Advanced Hybrid Deep Learning Model	89%	88%	88.5%	85%	This study
ConvNets [22]	92.51%	86.96%	-	-	External Benchmark
Deep Neural Networks [23]	75%	75%	-	-	External Benchmark
Structured Prediction [24]	88.12%	91.78%	-	-	External Benchmark
YOLOv5s [19]	51.2%	89.1%	-	-	External Benchmark

V. Conclusion And Future Scope

This study has successfully established a robust foundation for advancing the management of road infrastructure in Malaysia through the development and application of the Road Crack Dataset from International Islamic University Malaysia (RCD-IIUM). Employing innovative data collection and processing techniques, the study utilized high-resolution imaging, precise geospatial mapping, and sophisticated calibration methods to ensure comprehensive coverage and accuracy of the dataset. These efforts have resulted in a high-quality dataset that captures the intricacies of road pavement conditions across significant urban road networks in Selangor and Kuala Lumpur.

The importance and utility of the RCD-IIUM dataset extends beyond its immediate application in machine learning models. It serves as a crucial resource for urban planners and policymakers, providing a data-driven basis for enhancing road pavement maintenance strategies. The detailed insights derived from the dataset facilitate more informed decision-making, enabling the implementation of targeted maintenance that is both efficient and cost-effective. Furthermore, the dataset's potential to influence broader infrastructure policies cannot be overstated, as it supports the development of smarter, resilience-focused urban planning initiatives.

In terms of model application, the deployment of three advanced deep learning models—Customized YOLOv7, YOLOv8X-SEG, and an Advanced Hybrid Deep Learning Model—demonstrated exceptional capabilities in detecting and classifying various types of road pavement cracks. These models achieved high levels of precision, recall, and overall accuracy, underscoring their effectiveness in real-world applications. The successful application of these models highlights the dataset's robustness and confirms its potential as a benchmark tool for further research and development in the field of road condition assessment.

Looking forward, there are multiple pathways to extend the impact of this research:

- Enhancing Model Accuracy: Continuous refinement of the existing models could include integrating more varied data from different environmental conditions and road types to increase their generalizability and accuracy.
- Integration into Monitoring Systems: There is a significant opportunity to incorporate these models into real-time road monitoring systems, which would allow for ongoing assessments and timely interventions in road maintenance.
- Dataset Expansion: Expanding the dataset to include a wider range of geographical locations and road conditions will enhance its representativeness and utility, making it invaluable for national infrastructure management.
- Cross-Disciplinary Applications: The methodologies and technologies developed in this study have potential applications in other critical areas of public infrastructure, offering a model for similar enhancements in utility management and public building maintenance.

Accessibility Of The Dataset

The RCD-IIUM dataset is designed to be a valuable resource for both the academic community and professionals in road maintenance and infrastructure management. It is freely available on GitHub, ensuring easy accessibility for all users interested in utilizing it for research and practical applications. Copy the link below and paste it into your browser instead of clicking directly from this document.

GitHub Link: https://github.com/ArselanAshraf/RCD-IIUM-Road-Crack-Detection-Dataset-

To maintain the integrity and acknowledge the contribution of those who developed and curated the dataset, users are required to properly cite the relevant papers. The necessary citations are detailed in the README section of the GitHub repository. It is crucial that these references are included in any publications or presentations that utilize the dataset. This strategy aims to balance the need for open scientific collaboration with the recognition and integrity of the contributors' intellectual property.

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